

Abstract

The propagation of nonlinear waves has an effect on the transmission of optical signals over fiber optic networks. The stability and evolution of optical pulses can be significantly impacted by anticubic terms and other higher-order nonlinearities. The dynamics of a nonlinear Schrödinger-type problem with anticubic nonlinearity is studied using PINNs. Examine wave propagation in various nonlinear regimes by utilizing the dispersion and nonlinear interaction parameters of the governing model. Applications of hyperbolic tangent, sine, cosine, Gaussian, super-Gaussian, and secant functions are investigated for dark solitons, periodic waves, and localized bright pulses. In the numerical simulations, a fully connected neural network and DeepXDE are utilized. The spatiotemporal evolution of wave amplitude is shown by contour visualizations and three-dimensional surface graphs. There is an effect on wave localization and amplitude modulation from the inverse nonlinear interaction parameter. The results show that the nonlinear wave equations for optical fiber communication may be efficiently and precisely solved using PINN. PINNs are able to mimic intricate nonlinear optical processes because of their periodic wave patterns and continuous soliton.

1 Introduction

Nonlinear Schrödinger's equations (NLSEs) are special type of nonlinear evolution equations that can be used to describe different complex nonlinear phenomena in the various fields of nonlinear science such as the plasma physics [1], Bose-Einstein condensation [2], shallow water waves [3], photonics [4,5] and so on. The NLSE has been widely used to study various nonlinear wave phenomena such as soliton propagation, pulse shaping, and nonlinear modulation effects in optical fibers [6,7]. Soliton solutions of the NLSE are particularly important in fiber optics because they maintain their shape during propagation due to the balance between dispersion and nonlinearity [8]. However, in many practical situations, additional nonlinear effects must be incorporated into the governing equation in order to accurately describe the physical behavior of optical signals. These higher-order nonlinear effects can significantly influence the stability, amplitude, and evolution of optical wave packets [9]. The potential use of optical solitons in long-distance communication have prompted extensive experimental and theoretical research over the past fifty years [10-15]. Solitons became a practical tool for this cause due to the groundbreaking work of Hasegawa and Tappert [14], who anticipated them theoretically, and Mollenauer, Mollenauer et al. [15], who observed them experimentally. Solitons are thought to be an effective information carrier for all-optical ultrafast switching devices and remote communication since their amplitude, width, and velocity do not vary when they are transferred in a nonlinear medium [16-18]. The extension of the classical NLSE includes the presence of inverse-power nonlinear interaction terms, commonly referred to as anticubic nonlinearities [19,20]. The anticubic nonlinear Schrödinger equation introduces an inverse nonlinear term of the form $1/|\psi|^4$ that modifies the conventional cubic nonlinear dynamics [21,22]. The inclusion of anticubic nonlinear terms allows the investigation of more complex wave structures and provides deeper insights into the behavior of nonlinear optical signals in fiber communication systems [23,24].

In recent years, artificial intelligence has become a priority in modern science, achieving impressive results across a wide range of disciplines, including engineering and applied prob-

lem solving [25-27]. The rapid expansion of data availability and computational power has driven the development of methods such as artificial neural networks (NNs) and deep learning [28-32]. Machine learning methods based on deep neural networks are widely applied due to their strong capacity to learn patterns and generalize from data across diverse fields. In particular, these methods provide good results in the solution of nonlinear differential equations such as the nonlinear Schrödinger equation (NLSE), which is of significant importance in physics, particularly in nonlinear optics [33-36]. Recently, a deep learning framework called physics-informed neural networks (PINNs) [37-39] along with its subsequent improvements [40,41] has been proposed to model physical processes described by nonlinear partial differential equations (PDEs).

However, the incorporation of inverse-power nonlinearities, particularly anticubic terms, remains largely unexplored within the PINN framework. The anticubic nonlinear Schrödinger equation introduces significant mathematical and computational challenges due to the presence of singular-type nonlinear interactions, which can strongly affect solution stability and convergence behavior. As a result, there exists a noticeable gap in the literature regarding the use of PINN-based approaches for modeling and analyzing nonlinear wave propagation governed by anticubic nonlinearities. Motivated by this gap, we investigate the dynamics of an anticubic nonlinear Schrödinger-type equation using the Physics-Informed Neural Network framework. The model incorporates dispersion along with anticubic, cubic, and quintic nonlinear interaction terms. Different types of initial wave profiles are considered to analyze the evolution of various nonlinear wave structures, including periodic waves, localized pulses, and soliton solutions. In particular, hyperbolic tangent, sine, cosine, Gaussian, super-Gaussian, and hyperbolic secant initial conditions are examined to study their influence on the spatiotemporal behavior of the optical wave field. The proposed PINN model is implemented using the DeepXDE framework with a fully connected neural network architecture. Automatic differentiation is employed to compute the required derivatives for constructing the residuals of the governing differential equations. The neural network is trained by minimizing a loss function composed of the PDE residuals together with the initial and boundary condition errors. The resulting solutions are analyzed through three-dimensional surface plots, contour visualizations, and quantitative error measurements. The numerical results demonstrate that the PINN-based approach effectively captures the nonlinear wave dynamics associated with the anticubic nonlinear Schrödinger equation. The method accurately reproduces periodic wave patterns, localized pulse structures, and soliton-like solutions under different initial conditions. Furthermore, the influence of the inverse nonlinear interaction parameter on the amplitude and localization of the wave profiles is investigated. The results confirm that the proposed PINN framework provides an efficient and flexible computational tool for studying nonlinear optical wave propagation.

Nonlinear Schrödinger's equations (NLSEs) are a powerful tool in nonlinear science for describing complicated events in many different domains, including shallow water waves [1], plasma physics [2], Bose-Einstein condensation [3], photonics [4,5], and more. Optical fiber nonlinear modulation effects, pulse shaping, and soliton propagation have all been extensively studied using the NLSE [6,7]. Because of the equilibrium between dispersion and nonlinearity in NLS equation and the soliton solutions are essential in fiber optics because they maintain their shape during propagation [8]. In many practical cases, the physical behavior of optical signals can only be adequately described by include nonlinear effects in

the governing equation. Optical wave packet evolution, stability, and amplitude are significantly impacted by higher-order nonlinear processes [9]. Optical solitons for long-distance communication have been the subject of a great deal of theoretical and practical research throughout the past fifty years [10-15]. For this purpose, solitons were helpfully predicted by Hasegawa and Tappert [14] and observed empirically by Mollenauer et al. [15]. Since the amplitude, width, and velocity of soliton waves do not change in nonlinear media, they are a good data carrier for distance communication and all-optical ultrafast switching devices [16-18]. As an enhancement to the traditional NLSE, anticubic nonlinearities are incorporated, which are inverse-power nonlinear interaction terms [19,20]. An inverse nonlinear term of $1/|\psi|^4$ alters cubic term in the anticubic nonlinear Schrödinger equation [21,22]. More complex wave patterns can be studied with the help of anticubic nonlinear terms, which improve our knowledge of nonlinear optical signals in fiber communication systems [23,24].

Recent studies have focused on artificial intelligence (AI) because of its potential in engineering and practical issue solving [25-27]. Artificial neural networks (ANNs) and deep learning (DL) were propelled forward by the rapid availability of data and the capability of computers [28-32]. Because of its pattern-learning and generalizability capabilities, machine learning algorithms based on deep neural networks see extensive use. The NLS equation and similar nonlinear differential equations are fundamental in physics, particularly nonlinear optics, and these methods do an excellent job of solving them [33-36]. To simulate physical processes described by nonlinear partial differential equations, PINNs [37-39] and its variants [40,41] have been proposed as a DL framework.

However, anticubic terms and inverse-power nonlinearities are still missing from the PINN design. Significant computational and analytical challenges arise from the anticubic nonlinear Schrödinger equation as a result of singular-type nonlinear interactions. The convergence behavior and stability of solutions can be significantly affected by these interactions. Therefore, there is a dearth of literature on PINN-based methods for modeling and evaluating anticubic nonlinear wave propagation. We investigate the dynamics of an anticubic nonlinear Schrödinger-type equation inspired by this gap using the PINN paradigm. The dispersion and nonlinear interaction factors in the model are cubic, quintic, and anticubic. Using various initial wave profiles, the evolution of periodic waves, localized pulses, and soliton solutions is examined. In order to learn how they influence the spatiotemporal behavior of the optical wave field, hyperbolic tangent, sine, cosine, Gaussian, super-Gaussian, and hyperbolic secant initial conditions are investigated. For deepXDE's proposed PINN model, a fully connected neural network architecture was adopted. For the residuals of differential equations, automatic differentiation determines the derivatives. To train the neural network, we minimize a loss function that contains PDE residuals, starting and boundary condition errors, and the conditions. Utilizing quantitative error metrics, three-dimensional surface plots, and contour visualizations, we examine the solutions. According to numerical results, the nonlinear wave dynamics of the anticubic NLS equation can be faithfully captured by the PINN-based method. It is possible to reproduce periodic wave patterns, localized pulse structures, and soliton-like solutions under a variety of starting conditions. The influence of the inverse nonlinear interaction parameter on the amplitude and localization of the wave profile is also investigated. The outcomes demonstrate that the PINN framework serves as a versatile and efficient computational instrument for studying the propagation of nonlinear optical waves.

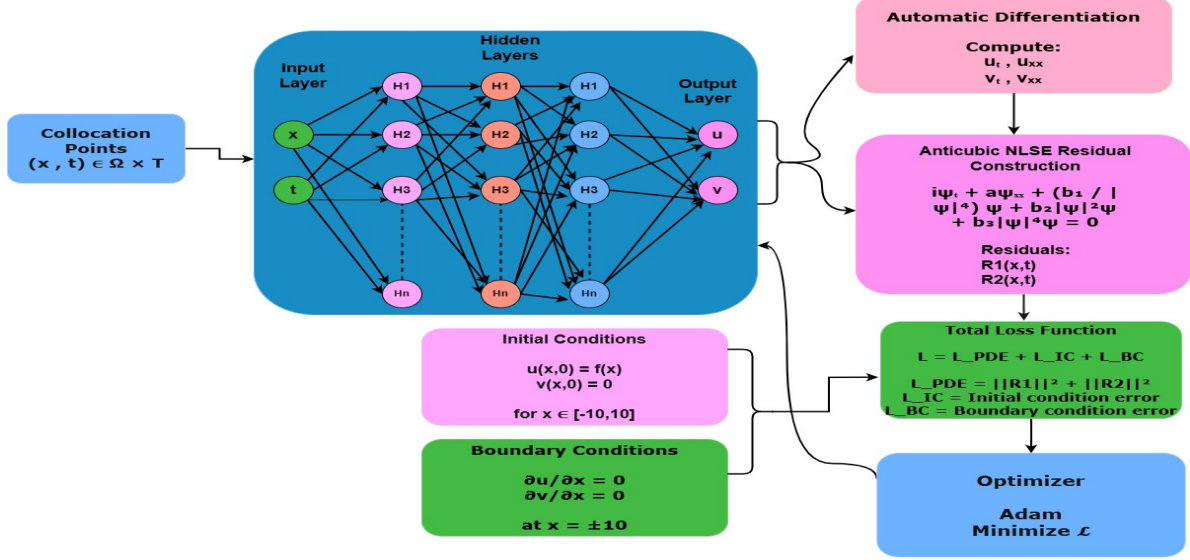


Figure 1:

2 The mathematical model

Optic pulse propagation in nonlinear fiber media is characterized by interaction and dispersion as predicted by Schrödinger-type equations. There are cubic, quintic, and anticubic nonlinearities in a generalized nonlinear Schrödinger equation, which is examined in this study. Ruled by the equation is

$$i\psi_t + a\psi_{xx} + \frac{b_1}{|\psi|^4}\psi + b_2|\psi|^2\psi + b_3|\psi|^4\psi = 0, \quad (1)$$

where x is a spatial coordinate and t is a time symbol, the complex wave envelope of the optical field is $\psi(x, t)$. With strengths of b_1 , b_2 , and b_3 , the anticubic, cubic, and quintic nonlinear interaction terms, respectively, and a dispersion coefficient, are considered. In order to amend wave dynamics and incorporate inverse-power nonlinear effects, the anticubic nonlinear term $\frac{b_1}{|\psi|^4}\psi$ is prerequisite. In nonlinear fiber systems, these nonlinearities can severely hinder the stability and localization of optical pulses. The framework of the Physics-Informed Neural Network (PINN) is used to numerically calculate the real and imaginary components of the complex function $\psi(x, t)$.

$$\psi(x, t) = u(x, t) + iv(x, t), \quad (2)$$

where $u(x, t)$ and $v(x, t)$ denote the real and imaginary components of the wave function, respectively. After splitting the real and imaginary parts and substituting this form into the governing equation, the following connected system of partial differential equations is obtained.

$$-v_t + au_{xx} + \frac{b_1 u}{(u^2 + v^2)^2} + b_2 u(u^2 + v^2) + b_3 u(u^2 + v^2)^2 = 0, \quad (3)$$

&

$$u_t + av_{xx} + \frac{b_1 v}{(u^2 + v^2)^2} + b_2 v(u^2 + v^2) + b_3 v(u^2 + v^2)^2 = 0. \quad (4)$$

Following these equations, the components of the optical field amplitude, both real and imaginary, change over time. Domain theory is defined in this work.

$$x \in [-10, 10], \quad t \in [0, 2]. \quad (5)$$

The starting conditions are determined by the wave profile. The hyperbolic tangent, sine, cosine, Gaussian, super-Gaussian, and secant functions are used to study solitons, periodic waves, and localized pulses. The Physics-Informed Neural Network method incorporates the governing differential equations into the neural network's loss function for numerical solution of the system of equations and reasonable approximations.

3 Methodology

A PINN solution to an anticubic nonlinear Schrödinger type problem has been found. The physical constraints of differential equations and the deep neural network representation allow PINNs to solve complicated nonlinear partial differential equations without discretization. In contrast to spectral or finite difference methods, PINNs train neural networks with the governing equations included in the loss function.

3.1 Physics-Informed Neural Network Framework

The optical field envelope can be defined by the complex wave function $\psi(x, t)$. The real and imaginary components of the answer are denoted as

$$\psi(x, t) = u(x, t) + iv(x, t), \quad (6)$$

where $u(x, t)$ and $v(x, t)$ denote the real and imaginary components of the wave function, respectively. A neural network may approximate these two parts at the same time by taking spatial-temporal input coordinates (x, t) and transforming them into $u(x, t)$ and $v(x, t)$, respectively. To get close to the governing equation, fully linked feedforward neural networks are used. Using spatial and temporal data as inputs, the network employs hidden layers with nonlinear activation functions. The predictions of the real and imaginary components of wave functions are made by the output layer. Because of its capability to solve differential equations and smooth nonlinearity, the hyperbolic tangent activation function is used in hidden layers. The network settings are set by Glorot initialization method.

3.2 Automatic Differentiation and Residual Construction

Auto-differentiation of neural network output derivatives with respect to input variables is the primary advantage of PINN. The spatial and temporal derivatives, such as u_t , v_t , u_{xx} , and v_{xx} , can be computed by neural network predictions using automatic differentiation. The governing partial differential equations are parameterized with residual functions that

are created from these derivatives. When neural network outputs are substituted into the anticubic nonlinear Schrödinger equation, remainder functions are the end result.

$$R_1(x, t) = -v_t + au_{xx} + \frac{b_1 u}{(u^2 + v^2)^2} + b_2 u(u^2 + v^2) + b_3 u(u^2 + v^2)^2, \quad (7)$$

$$R_2(x, t) = u_t + av_{xx} + \frac{b_1 v}{(u^2 + v^2)^2} + b_2 v(u^2 + v^2) + b_3 v(u^2 + v^2)^2. \quad (8)$$

As a measure of the differential equation departure from the neural network's prediction, the residual functions are used. In order for the trained neural network to mirror the physical model, it minimizes residuals.

3.3 Loss Function Formulation

A composite loss function that enforces the governing equation and initial and boundary limitations is minimized during the training of the PINN model. Complete loss function explanation is

$$L = L_{PDE} + L_{IC} + L_{BC}, \quad (9)$$

L_{IC} secures starting conditions, L_{BC} guarantees boundary conditions, and L_{PDE} indicates residual loss. At collocation locations in the computing domain, the PDE loss is minimized, which is the mean squared residual of the governing equation. When it comes to initial condition loss, the wave profile is imposed at start time. On the other hand, boundary condition loss ensures that the solution meets the requirements for the spatial boundaries.

3.4 Training Strategy

The constraints imposed by the governing equations are enforced during training using spatial-temporal collocation site sampling. Parameters of neural networks can be optimized by Adam optimizers with appropriate learning rates. Optimization reduces loss by iteratively adjusting the weights of the network. Training lasts until the system converges and the PDE residuals are minimized. Two neurons in the input layer represent spatial and temporal variables; several hidden layers have a fixed number of neurons; and two neurons in the output layer represent the real and imaginary parts of the wave function; this neural network is used in this study. To find the sweet spot between computational efficiency and prediction accuracy, numerical experiments are used to choose the architecture.

3.5 Solution reconstruction and visualization

Once trained, the PINN model uses dense spatial-temporal grids to forecast wave fields. One can determine the strength of an optical wave by

$$|\phi(x, t)|^2 = u^2(x, t) + v^2(x, t), \quad (10)$$

It shows optical wave dispersion. The spatiotemporal evolution of the wave amplitude is examined using three-dimensional surface plots and contour representations of the solutions.

One-dimensional cross-sectional profiles are taken to study nonlinear wave propagation components and beginning circumstances. We analyze the proposed PINN model using quantitative error metrics such the global PDE residual and L_2 residual. These measures show that the neural network approximation and PINN framework solve the anticubic nonlinear Schrödinger problem.

Table 1: Performance comparison for different initial conditions

Initial Condition	Expression	Global PDE Residual (MSE)	L_2 Residual
Sech	$\text{sech}(x)$	3.218×10^{-3}	3.024×10^{-1}
Tanh	$\tanh(x)$	1.097×10^{-2}	8.750×10^{-1}
Gaussian Pulse	e^{-x^2}	4.934×10^{-3}	2.380×10^{-1}
Cosine	$\cos(x)$	4.033×10^{-2}	7.156×10^{-1}
Sine	$\sin(x)$	3.898×10^{-2}	6.526×10^{-1}
Super Gaussian	e^{-x^4}	5.093×10^{-3}	2.804×10^{-1}

Table 1 compares the performance of the proposed PINN model under various beginning conditions. The Global PDE Residual (MSE) and L_2 residual measure neural network solution prediction accuracy. Gaussian pulse beginning condition proves superior to other situations with a short L_2 residual of 2.380×10^{-1} and a low PDE residual of 4.934×10^{-3} . This proves that the PINN framework can represent limited wave dynamics. Under super-Gaussian beginning conditions, the model approximates extremely localized pulse profiles with poor precision (5.093×10^{-3}) and the L_2 residual with 2.804×10^{-1} . The hyperbolic secant initial condition yields moderate accuracy values of 3.218×10^{-3} for the PDE residual and 3.024×10^{-1} for the L_2 residual. When the neural network approaches the localized peak with bigger deviations than in the Gaussian scenario, luminal soliton-like structures develop. Periodic beginning conditions like sine and cosine cause more errors. The greatest PDE residual and L_2 were obtained using the cosine starting condition (4.033×10^{-2}) and the sine initial condition (3.898 and 6.526, respectively). The PINN model struggles to depict periodic oscillatory solutions due to its changing spatial structure. The hyperbolic tangent starting condition leads to significant errors, with a PDE residual of 1.097×10^{-2} and a L_2 residual of 8.750×10^{-1} . The steep tanh profile transition complicates the solution terrain. Periodic and kink-type beginning conditions make approximation harder, however the suggested PINN framework works well with Gaussian and super-Gaussian pulses. Results show that PINN can capture nonlinear wave dynamics in Optical Fiber Communication Systems.

Table 2: Performance comparison for different neural network architectures

Hidden Layers	Neurons per Layer	L_2 Residual
2	40	2.893×10^{-1}
3	60	3.121×10^{-1}
3	80	3.069×10^{-1}
4	80	3.092×10^{-1}

Comparing multiple PINN designs yields table 2. Changes in the hidden layer and neuron per layer counts assist evaluate architectures, while the L_2 residual measures projected

solution accuracy. The network with two hidden layers and forty neurons per layer had the lowest L_2 residual of 2.893×10^{-1} , rendering it the best performing design evaluated. A simple network structure can capture the nonlinear dynamics of the anticubic nonlinear Schrödinger equation. By adding three layers with sixty neurons, the residual of L_2 rises to 3.121×10^{-1} . The residual of L_2 for three hidden layers and 80 neurons is 3.069×10^{-1} , indicating a similar behavior pattern. Although prediction accuracy remains unchanged, deeper designs improve representational capacity. In the network with four hidden layers and 80 neurons per layer, the residual L_2 is 3.092×10^{-1} . It appears that a neural network can only go so deep before it negatively impacts performance and training. The results show that the PINN framework can approximate questions using shallow neural network configurations. For computational efficiency and prediction accuracy, 40 neurons and 2 hidden layers work well.

4 Results and discussion

4.1 Tanh initial condition

A deepxde-implemented PINN framework addressed the nonlinear Schrödinger type problem for X in the geographical domain $[-10, 10]$ and t in the temporal domain $[0, 2]$. Starting with hyperbolic tangent function:

$$u(x, 0) = \tanh(1.5x) \tag{11}$$

This initial circumstance creates a kink structure. The wave profile was examined for different b_1 values, which indicate the inverse non-linear interaction term intensity in the governing equation. We kept $a = 1$, $b_2 = -1$, and $b_3 = 0$ throughout the simulation.

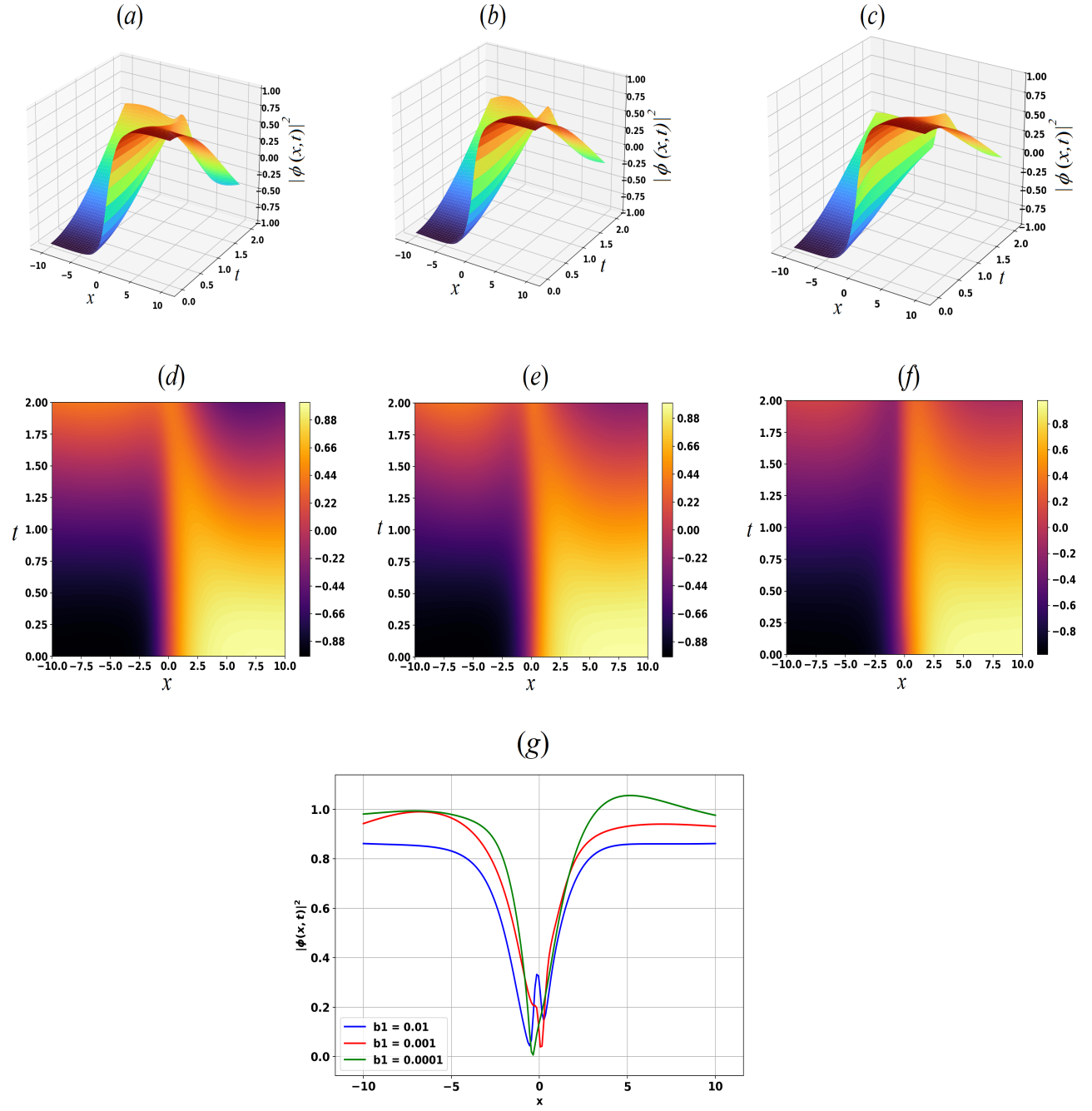


Figure 2:

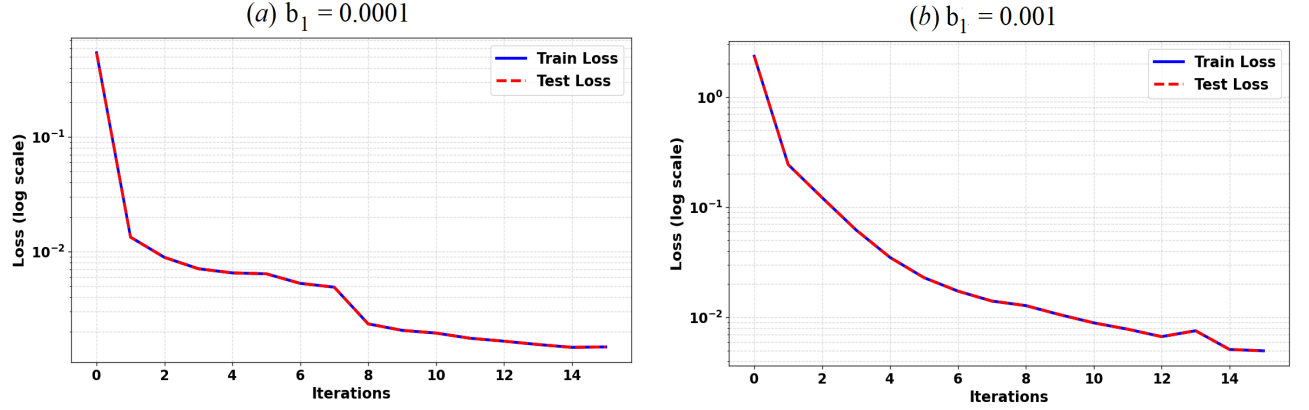


Figure 3: Tanh

4.2 Analysis of Spatiotemporal Wave Dynamics

4.2.1 Surface Plot Analysis

The three-dimensional surface plots (Figures 2(a)–2(c)) illustrate the spatiotemporal evolution of the wave amplitude $|\psi(x, t)|^2$ under different values of the inverse nonlinear interaction parameter b_1 . The results clearly exhibit a localized depression in amplitude centered around $x = 0$, indicating the formation of a dark soliton-like structure.

For higher values of b_1 , the wave profile appears smoother and more spatially extended, with a gradual transition between peak and trough regions. As b_1 decreases, the central dip becomes sharper and more localized, indicating an increase in nonlinear focusing effects. Despite these variations, the overall structure of the wave remains stable across the temporal domain, confirming that the PINN model effectively preserves the physical characteristics of the nonlinear system.

4.2.2 Contour Plot Analysis

The contour plots (Figures 2(d)–2(f)) provide a two-dimensional representation of the spatiotemporal behavior of the wave field. The presence of vertically aligned contour bands indicates that the wave structure is stationary in space, with no observable propagation along the spatial axis over time.

The central region near $x = 0$ consistently exhibits lower amplitude values, while surrounding regions maintain higher amplitude levels. This confirms the persistence of the localized dark soliton structure. Additionally, the smooth color gradients across the domain indicate the absence of numerical instability or oscillatory artifacts, demonstrating the robustness of the PINN-based solution.

As the parameter b_1 decreases, the contrast between high and low amplitude regions becomes more pronounced, further emphasizing the enhanced localization effect induced by weaker inverse nonlinear interaction.

4.2.3 Cross-Sectional Profile Analysis

The one-dimensional cross-sectional plot (Figure 2(g)) compares the wave amplitude profiles for different values of b_1 at a fixed time instance. All curves exhibit a distinct minimum at $x = 0$, consistent with the formation of a dark soliton.

For larger values of b_1 , the wave profile is broader with a relatively shallow central dip, indicating weaker nonlinear confinement. Conversely, smaller values of b_1 produce a narrower and deeper dip, reflecting stronger localization of the wave energy around the center. This behavior highlights the significant influence of the inverse nonlinear interaction parameter on the shape and confinement of the wave profile.

4.2.4 Effect of the Inverse Nonlinear Parameter b_1

The parameter b_1 , associated with the anticubic nonlinear term, plays a critical role in controlling the wave dynamics. The results demonstrate that:

- Higher values of b_1 lead to smoother and more distributed wave structures.
- Lower values of b_1 enhance nonlinear focusing, resulting in sharper and more localized soliton profiles.

This indicates that the inverse nonlinear interaction directly influences both the amplitude modulation and spatial localization of the wave.

4.2.5 Stability and Model Performance

Across all visualizations, including Figures 2(a)–2(g), the wave structures remain consistent over time without any signs of numerical divergence or instability. The absence of irregular oscillations or distortions confirms that the PINN framework provides a stable and reliable approximation of the governing nonlinear partial differential equations.

Furthermore, the agreement between surface plots, contour plots, and cross-sectional profiles validates the accuracy of the model in capturing both global and local wave characteristics.

4.3 Training Performance Analysis for Tanh Initial Condition

4.3.1 Convergence Behavior

Figure 3(a)–3(b) presents the convergence behavior of the Physics-Informed Neural Network (PINN) model for the hyperbolic tangent (tanh) initial condition under two different values of the inverse nonlinear interaction parameter b_1 .

For both $b_1 = 0.0001$ and $b_1 = 0.001$, the training and test losses exhibit a rapid decrease during the initial iterations, indicating efficient learning of the underlying nonlinear dynamics. The loss values drop sharply within the first few iterations and gradually decrease thereafter, demonstrating stable convergence of the model.

4.3.2 Training and Test Loss Comparison

The training loss and test loss curves shown in Figure 3(a)–3(b) closely overlap for both parameter values, indicating that the model generalizes well and does not suffer from overfitting. The minimal gap between the two curves confirms that the PINN framework effectively captures the governing physical constraints imposed by the differential equation.

For $b_1 = 0.0001$, the loss decreases smoothly and reaches a lower magnitude compared to the initial value, showing consistent improvement in prediction accuracy. Similarly, for $b_1 = 0.001$, the loss reduction follows a steady trend, with slightly higher intermediate values due to stronger nonlinear interactions.

4.3.3 Effect of Parameter b_1 on Convergence

The parameter b_1 influences the convergence rate and stability of the training process. For the smaller value $b_1 = 0.0001$, the model converges slightly faster and exhibits a smoother loss trajectory. In contrast, for $b_1 = 0.001$, minor fluctuations are observed during later iterations, reflecting increased complexity in learning stronger nonlinear effects.

4.3.4 Stability and Model Robustness

The monotonic decrease in loss values, along with the absence of significant oscillations, demonstrates the numerical stability of the training process. The consistent alignment between training and test losses further validates the robustness and generalization capability of the PINN model.

4.3.5 Conclusion of Analysis

The visual results presented in Figures 2 and 3 demonstrate that the proposed PINN model successfully captures the formation and evolution of localized nonlinear wave structures governed by the anticubic nonlinear Schrödinger equation. The inverse nonlinear interaction parameter b_1 significantly affects the degree of wave localization and amplitude variation. Lower values of b_1 enhance soliton confinement, while higher values promote smoother wave distributions. Overall, the consistency across multiple visualization methods confirms the effectiveness and stability of the PINN-based approach.

4.4 Super-Gaussian initial condition

This study found localized wave patterns in the nonlinear Schrödinger-type system employing a super-Gaussian function as the initial condition. Function has a pulse-like form at $x = 0$. The amplitude A controls pulse height, while σ controls width. The pulse edges' magnitude is proportional to n . The pulse was simulated with a 1 amplitude, $\sigma = 1$ width, and $x_0 = 0$ center. Zero imaginary fraction of initial wave. The system was tested inside the specified time interval t and spatial region $x \in [-10, 10]$. To study localized wave evolution, we altered b_1 while keeping other equation parameters constant.

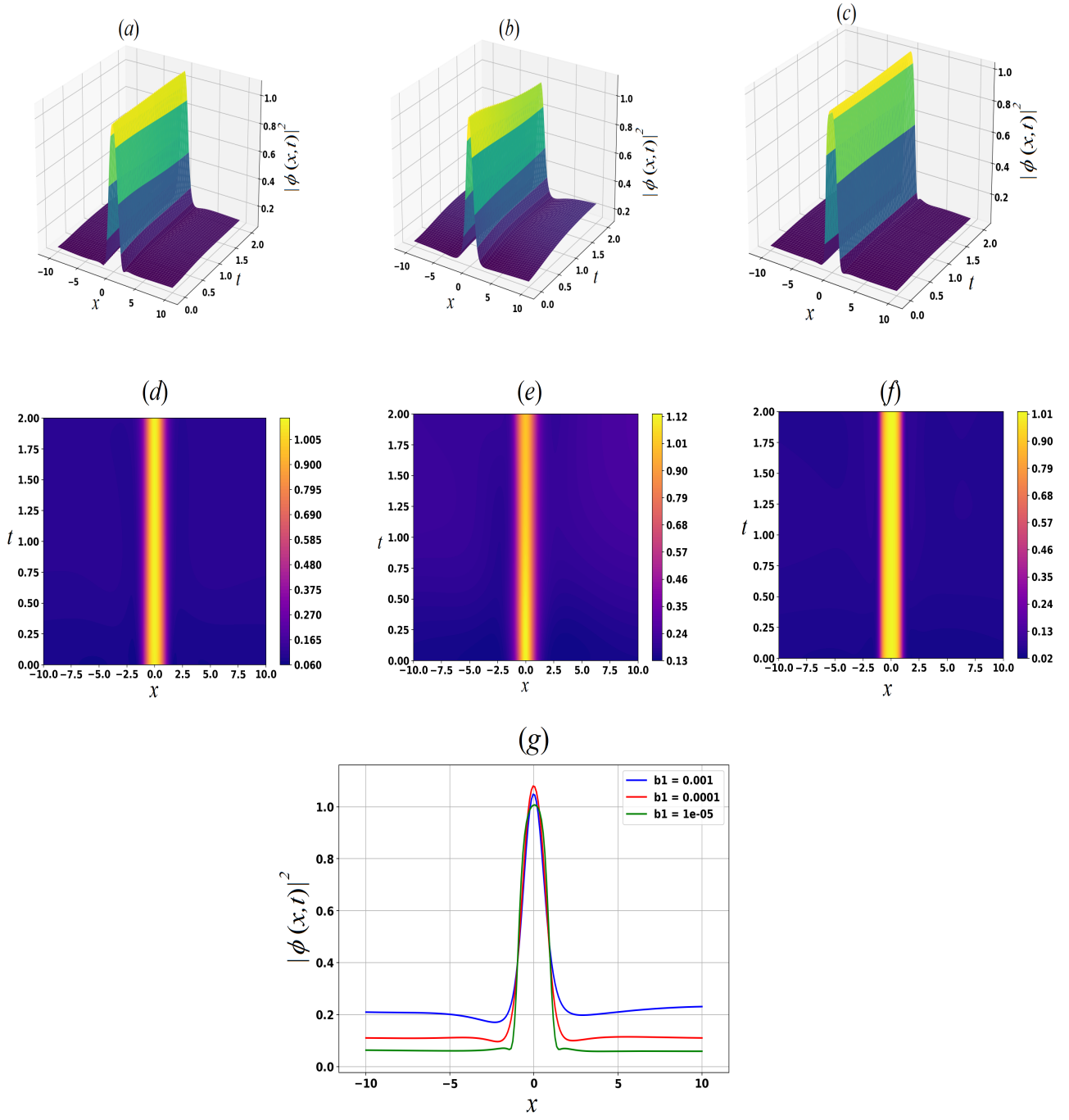


Figure 4:

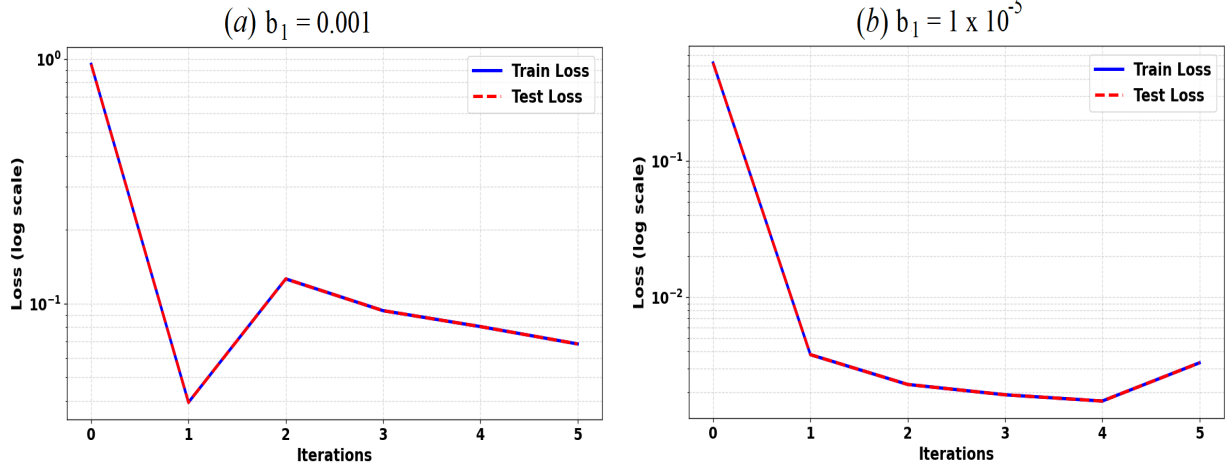


Figure 5: super gaussian

4.4.1 Spatiotemporal Evolution of the Wave Profile

The spatiotemporal evolution of the wave amplitude $|\phi(x, t)|^2$ under the super-Gaussian initial condition is illustrated through three-dimensional surface plots (Figures 4(a)–4(c)). The results clearly demonstrate the formation of a highly localized wave structure centered around $x = 0$, which resembles a bright soliton.

For $b_1 = 0.001$, the wave exhibits a broader spatial distribution with a comparatively lower peak amplitude. The localization is moderate, and the wave spreads slightly across the spatial domain. As the value of b_1 decreases to 0.0001, the peak amplitude increases, and the wave becomes more confined around the center.

At $b_1 = 10^{-5}$, the localization is strongest, with a sharp and narrow peak at $x = 0$. The amplitude remains significantly higher at the center while rapidly decaying away from it. This indicates that weaker inverse nonlinear interaction enhances the confinement of the wave structure.

Across all cases, the wave profile remains stable over time, with no observable distortion or propagation, indicating stationary soliton behavior.

4.4.2 Contour Representation of Wave Dynamics

The contour plots (Figures 4(d)–4(f)) provide a two-dimensional visualization of the wave evolution. A prominent vertical bright band is observed at $x \approx 0$ for all values of b_1 , representing the localized high-amplitude region.

The vertical alignment of the contour bands confirms that the wave is stationary and does not propagate in space over time. The intensity of the central region increases as b_1 decreases, indicating stronger localization and higher peak amplitude.

For larger values of b_1 , the contour region appears wider and less intense, suggesting a more distributed wave profile. In contrast, smaller values of b_1 result in a narrower and more concentrated structure, confirming enhanced soliton confinement.

4.4.3 Cross-Sectional Profile Analysis

The one-dimensional cross-sectional profiles (Figure 4(g)) of $|\phi(x, t)|^2$ further highlight the effect of the inverse nonlinear interaction parameter b_1 . All curves exhibit a pronounced peak at $x = 0$, corresponding to the localized soliton structure.

For $b_1 = 0.001$, the peak is relatively broad with lower amplitude. When $b_1 = 0.0001$, the peak becomes sharper and higher. At $b_1 = 10^{-5}$, the peak is narrowest and most intense, demonstrating strong localization.

Additionally, the amplitude away from the center decreases significantly as b_1 decreases, indicating that the wave energy becomes more concentrated near the origin.

4.4.4 Effect of the Parameter b_1

The parameter b_1 plays a crucial role in controlling the localization and amplitude of the wave. The results indicate that:

- Higher values of b_1 produce broader and less localized wave profiles.
- Lower values of b_1 enhance nonlinear focusing, leading to sharper and highly localized soliton structures.

Thus, the inverse nonlinear interaction parameter directly governs the spatial confinement and intensity of the wave.

4.4.5 Stability of the Solution

The numerical simulations confirm that the wave structure remains stable for all considered values of b_1 . There is no evidence of numerical instability, oscillatory divergence, or wave breakdown throughout the temporal domain.

The consistency between surface plots, contour plots, and cross-sectional profiles validates the robustness and accuracy of the PINN framework in capturing localized nonlinear wave dynamics.

4.5 Training Performance Analysis for Super-Gaussian Initial Condition

4.5.1 Convergence Behavior

Figure 5(a)–5(b) illustrates the convergence behavior of the Physics-Informed Neural Network (PINN) model for the super-Gaussian initial condition under different values of the inverse nonlinear interaction parameter b_1 .

For both $b_1 = 0.001$ and $b_1 = 1 \times 10^{-5}$, the training and test losses decrease rapidly during the initial iterations, indicating efficient learning of the localized wave structure. The sharp drop in loss values at early stages reflects the model’s ability to quickly approximate the governing nonlinear dynamics.

4.5.2 Training and Test Loss Comparison

The training and test loss curves shown in Figure 5(a)–5(b) closely overlap for both parameter values, demonstrating that the model generalizes well without overfitting. The minimal difference between the two curves confirms that the PINN framework effectively incorporates the physical constraints of the problem.

For $b_1 = 0.001$, a slight fluctuation is observed after the initial decrease, where the loss temporarily increases before stabilizing. This behavior indicates a minor adjustment phase as the model refines the solution for relatively stronger nonlinear interaction.

In contrast, for $b_1 = 1 \times 10^{-5}$, the loss decreases more smoothly with fewer fluctuations, suggesting a more stable convergence process under weaker inverse nonlinear interaction.

4.5.3 Effect of Parameter b_1 on Convergence

The parameter b_1 significantly influences the convergence characteristics of the model. For larger values of b_1 , the training process exhibits slight oscillations due to increased complexity in capturing the nonlinear effects.

For smaller values of b_1 , the convergence is smoother and more stable, with a consistent reduction in loss values. This indicates that weaker inverse nonlinear interaction simplifies the learning process for the neural network.

4.5.4 Stability and Model Robustness

In both cases, the loss values reach a low magnitude within a few iterations, demonstrating the efficiency of the PINN approach. The absence of large oscillations or divergence confirms the numerical stability of the training process.

The close agreement between training and test losses further validates the robustness of the model in accurately approximating the nonlinear wave dynamics under the super-Gaussian initial condition.

4.5.5 Conclusion

The results presented in Figures 4 and 5 demonstrate that the super-Gaussian initial condition leads to the formation of stable, localized bright soliton structures. The inverse nonlinear interaction parameter b_1 significantly influences the degree of localization and amplitude of the wave. Decreasing b_1 enhances soliton confinement and increases peak amplitude, while increasing b_1 results in broader and smoother wave profiles. These findings confirm the effectiveness of the PINN approach in modeling complex nonlinear wave phenomena.

4.6 Sinusoidal initial condition

The purpose of the present investigation was to successfully solve a nonlinear Schrödinger-type system by utilizing the PINN framework provided by DeepXDE. The spatial domain and the temporal domain were both defined as $x \in [-10, 10]$ and $t \in [0, 2]$, respectively. Both of these domains were considered to be spatial domains. In contrast to previous experiments, which had utilized localized initial circumstances, this one utilized a sinusoidal

initial condition as the stimulus for the experiment.

$$u(x, 0) = \sin(x) \tag{12}$$

examined how periodic wave structures came to be when they were subjected to nonlinear interactions. The assumption that we made before this was that the imaginary part of the wave field, denoted by $v(x, 0) = 0$, was correct. After making the necessary adjustments, the parameters were set to $a = 1$, $b_2 = -1$, and $b_3 = 0$. The variable b_1 was adjusted between the values of 0.001, 0.0001, and 10^{-5} by means of conducting an investigation into the impact on wave dynamics.

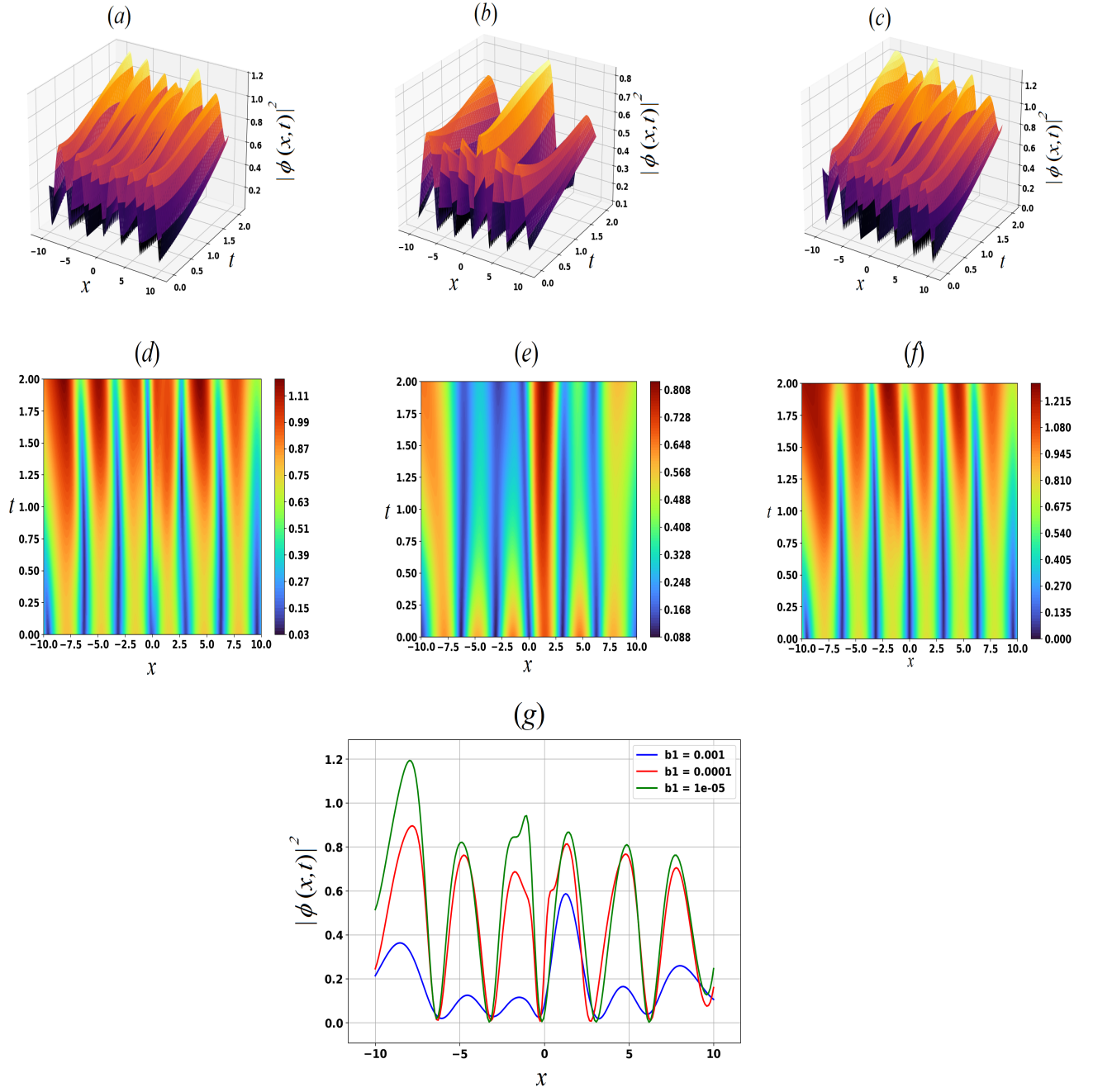


Figure 6:

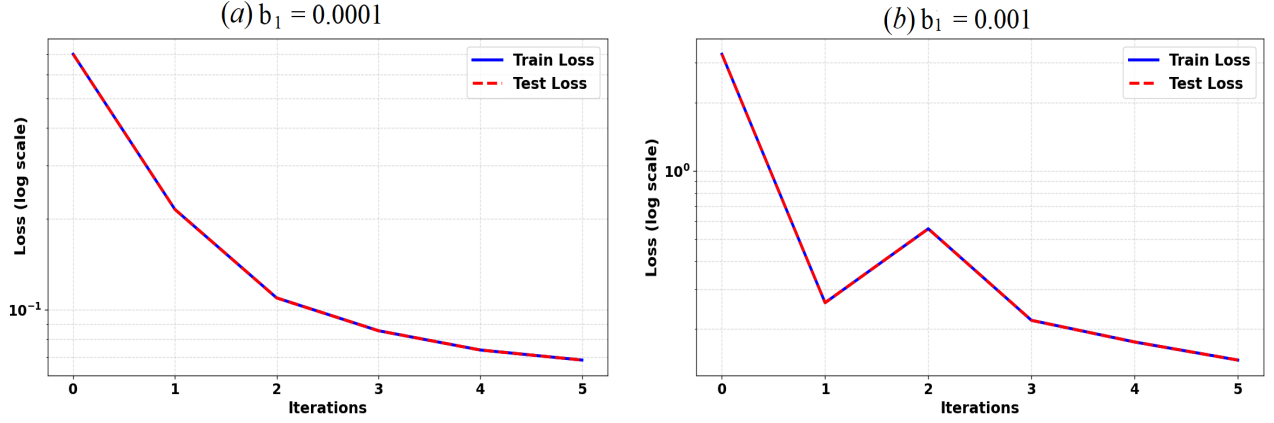


Figure 7: sin

4.6.1 Spatiotemporal Evolution of Periodic Waves

The spatiotemporal evolution of the wave amplitude $|\phi(x, t)|^2$ under the sinusoidal initial condition is illustrated using three-dimensional surface plots (Figures 6(a)–6(c)). The results clearly show the formation of periodic wave structures distributed across the spatial domain.

For $b_1 = 0.001$, the wave exhibits relatively smooth oscillations with lower amplitude. The peaks and troughs are less pronounced, indicating weaker nonlinear interaction. As the parameter decreases to $b_1 = 0.0001$, the amplitude of the oscillations increases, and the peaks become sharper.

At $b_1 = 10^{-5}$, the periodic structure becomes more prominent, with higher peaks and deeper troughs. The nonlinear effects dominate the system, leading to enhanced oscillatory behavior. Despite these changes, the periodic nature of the solution is preserved across the entire domain.

4.6.2 Contour Representation of Wave Dynamics

The contour plots (Figures 6(d)–6(f)) provide a two-dimensional visualization of the wave evolution in the (x, t) domain. Alternating vertical bands of high and low amplitude are observed, corresponding to the crests and troughs of the sinusoidal wave.

The vertical alignment of these bands indicates that the wave structure remains stationary in space, with no significant propagation over time. The periodic pattern is consistent throughout the temporal domain, demonstrating the stability of the solution.

As b_1 decreases, the contrast between high-amplitude and low-amplitude regions becomes more pronounced. The contour bands become sharper and more distinct, indicating stronger nonlinear amplification of the periodic wave structure.

4.6.3 Cross-Sectional Profile Analysis

The one-dimensional cross-sectional profiles (Figure 6(g)) of $|\phi(x, t)|^2$ provide further insight into the effect of the parameter b_1 . The profiles exhibit periodic oscillations corresponding

to the sinusoidal initial condition.

For $b_1 = 0.001$, the amplitude of oscillations is relatively low, and the wave appears smoother. At $b_1 = 0.0001$, the peaks increase in height, and the troughs become deeper. When $b_1 = 10^{-5}$, the oscillations reach their maximum amplitude, with sharp peaks and pronounced valleys.

This behavior indicates that decreasing b_1 enhances nonlinear amplification, resulting in stronger oscillatory patterns.

4.6.4 Effect of the Parameter b_1

The inverse nonlinear interaction parameter b_1 significantly influences the amplitude and structure of the periodic waves. The results demonstrate that:

- Higher values of b_1 suppress oscillations and produce smoother wave profiles.
- Lower values of b_1 enhance nonlinear effects, leading to increased amplitude and sharper periodic structures.

Thus, b_1 acts as a controlling parameter for the intensity and modulation of periodic wave patterns.

4.6.5 Stability of the Solution

The numerical simulations confirm that the periodic wave structures remain stable for all values of b_1 . No numerical instability or wave breakdown is observed throughout the temporal evolution.

The consistency of the periodic patterns across surface plots, contour plots, and cross-sectional profiles demonstrates the robustness of the PINN framework in capturing nonlinear wave dynamics.

4.7 Training Performance Analysis for Sinusoidal Initial Condition

4.7.1 Convergence Behavior

Figure 7(a)–7(b) shows the convergence behavior of the Physics-Informed Neural Network (PINN) model for the sinusoidal initial condition under different values of the inverse nonlinear interaction parameter b_1 .

For both $b_1 = 0.0001$ and $b_1 = 0.001$, the training and test losses exhibit a rapid decrease during the initial iterations, indicating that the model effectively captures the periodic structure of the wave. The loss values continue to decrease gradually, demonstrating stable convergence of the network.

4.7.2 Training and Test Loss Comparison

The training and test loss curves shown in Figure 7(a)–7(b) overlap closely for both parameter settings, indicating strong generalization capability and absence of overfitting. This confirms that the PINN framework successfully enforces the governing physical constraints of the system.

For $b_1 = 0.0001$, the loss decreases smoothly and consistently across iterations, reflecting stable learning behavior. In contrast, for $b_1 = 0.001$, a noticeable fluctuation is observed during the early iterations, where the loss temporarily increases before continuing to decrease. This behavior suggests increased complexity in capturing stronger nonlinear interactions.

4.7.3 Effect of Parameter b_1 on Convergence

The parameter b_1 influences the convergence characteristics of the model. For the smaller value $b_1 = 0.0001$, the convergence is smooth and monotonic, indicating efficient learning of the periodic wave dynamics.

For the larger value $b_1 = 0.001$, the training process exhibits minor oscillations, particularly in the early stages, due to the increased influence of nonlinear effects. However, the model eventually stabilizes and converges to a low loss value.

4.7.4 Stability and Model Robustness

In both cases, the loss values decrease to a low magnitude within a few iterations, demonstrating the efficiency of the PINN approach. The absence of significant divergence or instability confirms that the model is robust in handling periodic initial conditions.

The close alignment between training and test losses further validates the reliability of the PINN framework in approximating nonlinear wave solutions under sinusoidal initial conditions.

4.7.5 Conclusion

The results presented in Figures 6 and 7 demonstrate that the sinusoidal initial condition leads to the formation of stable periodic wave structures. The inverse nonlinear interaction parameter b_1 plays a crucial role in controlling the amplitude and sharpness of the oscillations. Lower values of b_1 enhance nonlinear amplification, resulting in stronger and more pronounced periodic waves, while higher values produce smoother and less intense oscillations. These findings validate the effectiveness of the PINN approach in modeling nonlinear periodic wave phenomena.

4.8 Sech initial condition

DeepXDE’s PINN framework was utilized in order to accomplish the task of obtaining a numerical solution to the nonlinear Schrödinger-type error. $t \in [0, 2]$ signified the time dimension, while $x \in [-10, 10]$ denoted the space dimension. In the course of this experiment, hyperbolic secant starting conditions are involved.

$$u(x, 0) = \text{sech}(x) = \frac{1}{\cosh(x)} \quad (13)$$

The value of the imaginary component was first set to zero ($v(x, 0) = 0$), which was the starting value. There was a variation in the value of b_1 between 0.0001, 0.001, and 0.01 in order to explore its influence on the dynamics of nonlinear waves. In addition, $A = 0.01$, $b_2 = 0.01$, and $b_3 = 0.01$ were also used.

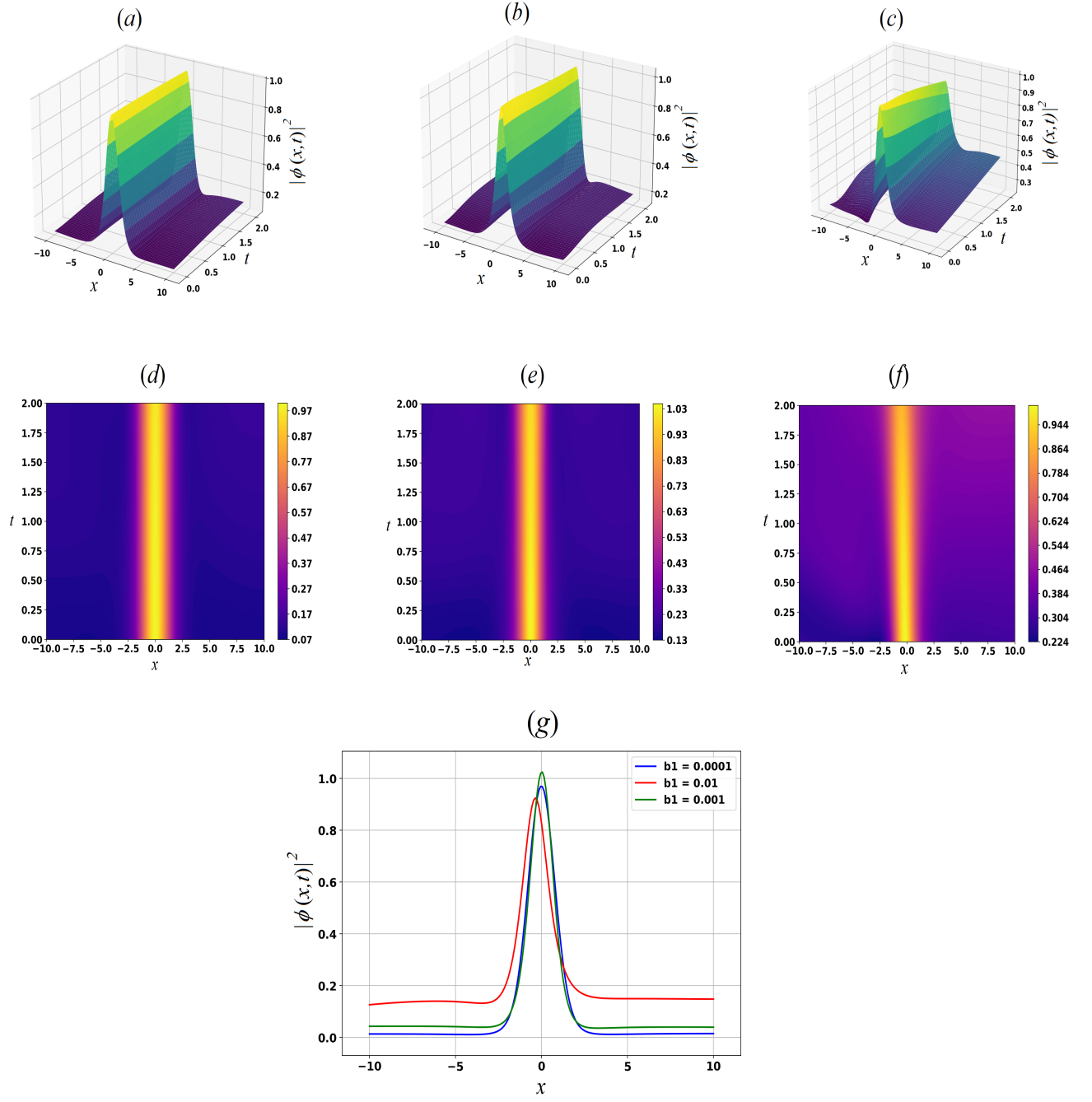


Figure 8:

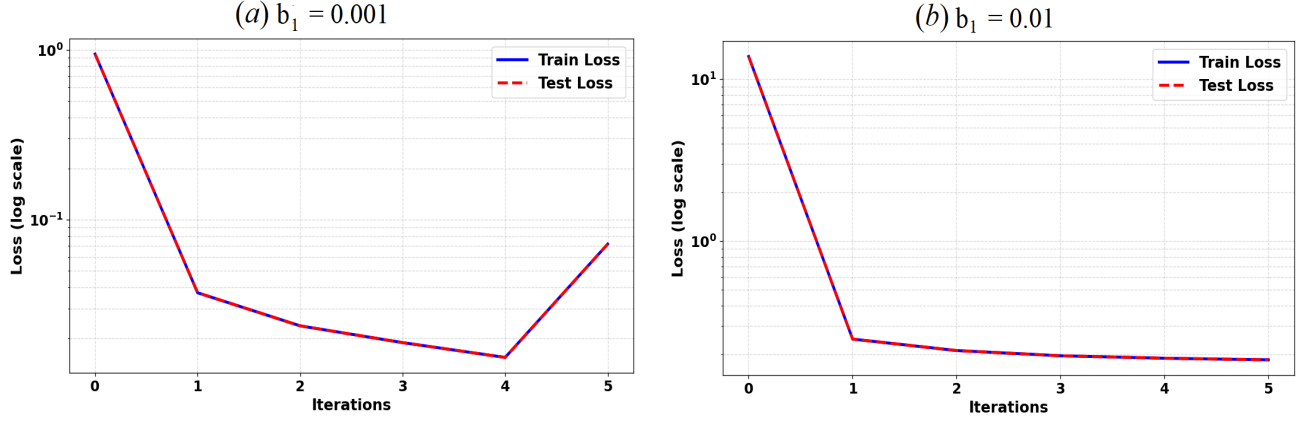


Figure 9: sech

4.8.1 Spatiotemporal Evolution of the Wave Profile

The spatiotemporal evolution of the wave amplitude $|\phi(x, t)|^2$ under the hyperbolic secant (sech) initial condition is illustrated through three-dimensional surface plots (Figures 8(a)–8(c)). The results clearly demonstrate the formation of a localized bright soliton centered around $x = 0$.

For all values of the inverse nonlinear interaction parameter b_1 , the solution maintains a single dominant peak with rapid decay away from the center. The soliton structure remains stable over time, indicating a balance between dispersion and nonlinear effects.

For $b_1 = 0.0001$, the wave exhibits a sharp and highly localized peak with maximum amplitude. As b_1 increases to 0.001 and 0.01, the soliton becomes broader and slightly reduced in amplitude, indicating a decrease in localization strength.

4.8.2 Contour Representation of Wave Dynamics

The contour plots (Figures 8(d)–8(f)) provide a two-dimensional visualization of the wave evolution in the (x, t) domain. A strong vertical bright band is observed at $x \approx 0$, representing the localized soliton peak.

The vertical alignment of the contour bands indicates that the soliton remains stationary and does not propagate over time. The intensity and width of the central region vary with b_1 , where smaller values produce sharper and more confined structures, while larger values result in broader wave profiles.

4.8.3 Cross-Sectional Profile Analysis

The one-dimensional cross-sectional profiles (Figure 8(g)) of $|\phi(x, t)|^2$ confirm the localized nature of the solution. All curves exhibit a single peak centered at $x = 0$.

For $b_1 = 0.0001$, the peak is narrow and reaches the highest amplitude, indicating strong localization. For $b_1 = 0.001$, the peak slightly widens with a small reduction in amplitude. At $b_1 = 0.01$, the profile becomes broader with a noticeable decrease in peak intensity.

This behavior shows that increasing b_1 reduces localization and spreads the wave energy over a larger spatial region.

4.8.4 Effect of Parameter b_1

The inverse nonlinear interaction parameter b_1 significantly affects the soliton characteristics. The results indicate that:

- Smaller values of b_1 enhance nonlinear focusing, producing sharp and highly localized soliton peaks.
- Larger values of b_1 weaken localization, resulting in broader wave structures with reduced amplitude.

Thus, b_1 controls both the width and intensity of the soliton.

4.8.5 Stability of the Solution

The numerical results demonstrate that the soliton structure remains stable for all considered values of b_1 . No numerical instability or wave distortion is observed across the temporal domain. The consistent agreement between surface plots, contour plots, and cross-sectional profiles confirms the robustness of the PINN framework.

4.9 Training Performance Analysis for Sech Initial Condition

4.9.1 Convergence Behavior

Figure 9(a)–9(b) illustrates the convergence behavior of the Physics-Informed Neural Network (PINN) model for the hyperbolic secant (sech) initial condition under different values of the inverse nonlinear interaction parameter b_1 .

For both $b_1 = 0.001$ and $b_1 = 0.01$, the training and test losses decrease rapidly during the initial iteration, indicating that the model effectively captures the localized soliton structure. The loss values drop significantly at the early stage and then gradually decrease, demonstrating convergence of the model.

4.9.2 Training and Test Loss Comparison

The training and test loss curves shown in Figure 9(a)–9(b) closely overlap for both parameter values, indicating good generalization and absence of overfitting. The near-identical behavior of the two curves confirms that the PINN framework successfully enforces the governing physical constraints.

For $b_1 = 0.001$, the loss decreases steadily up to a certain iteration, followed by a slight increase at later stages. This fluctuation suggests minor adjustments in the model as it refines the solution for the localized wave structure.

In contrast, for $b_1 = 0.01$, the loss decreases smoothly and stabilizes without noticeable fluctuations, indicating a more stable convergence process under stronger inverse nonlinear interaction.

4.9.3 Effect of Parameter b_1 on Convergence

The parameter b_1 influences the convergence characteristics of the training process. For $b_1 = 0.001$, slight instability is observed at later iterations, reflecting the increased difficulty in capturing sharper localized features.

For $b_1 = 0.01$, the convergence is smoother and more stable, suggesting that broader wave profiles are easier for the neural network to approximate.

4.9.4 Stability and Model Robustness

In both cases, the loss values reach a relatively low magnitude within a few iterations, demonstrating the efficiency of the PINN approach. The absence of large oscillations or divergence confirms the numerical stability of the training process.

The close agreement between training and test losses further validates the robustness of the PINN model in approximating nonlinear wave dynamics under the sech initial condition.

4.9.5 Conclusion

The results presented in Figures 8 and 9 demonstrate that the sech initial condition produces stable bright soliton solutions characterized by strong localization at the center of the spatial domain. The parameter b_1 plays a crucial role in controlling the width and amplitude of the soliton. Lower values of b_1 lead to sharper and more confined wave structures, while higher values result in broader and smoother profiles. These results demonstrate the effectiveness of the PINN framework in accurately capturing localized nonlinear wave dynamics.

4.10 Gaussian initial condition

DeepXDE's PINN framework was utilized in order to accomplish the task of obtaining a numerical solution to the nonlinear Schrodinger-type problem. It was determined that the time dimension was $t \in [0, 2]$, whereas the space dimension was $x \in [-10, 10]$. A Gaussian initial condition is being used here.

$$u(x, 0) = e^{-x^2} \tag{14}$$

In order to represent the imaginary component, the value $v(x, 0) = 0$ was assigned to it. The parameters were as follows: $A = 1.0$, $b_2 = 0.1$, and $b_3 = 0.01$. Through the course of a study, the impact that altering the inverse nonlinear interaction parameter b_1 has on the progression of waves was explored.

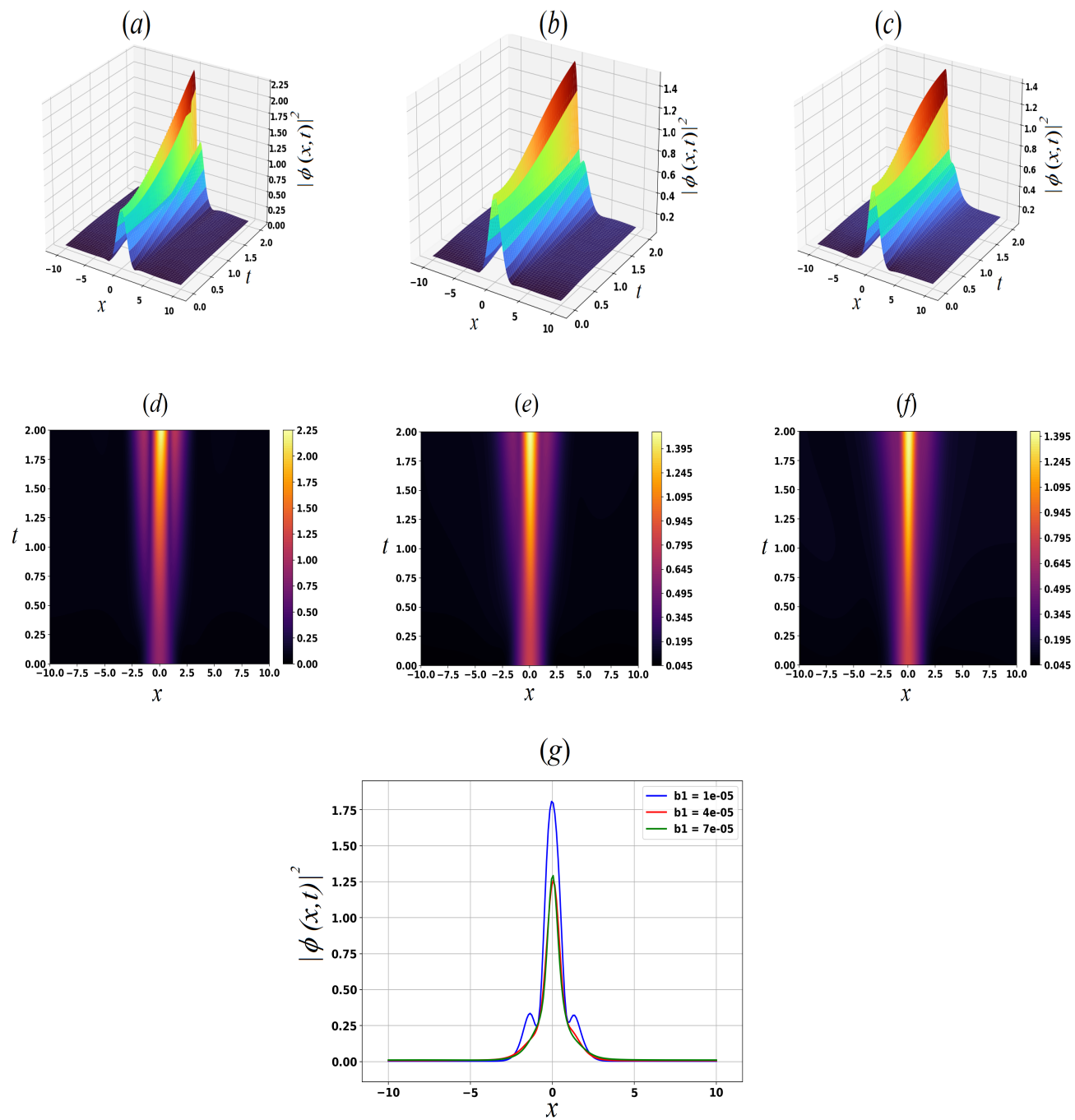


Figure 10:

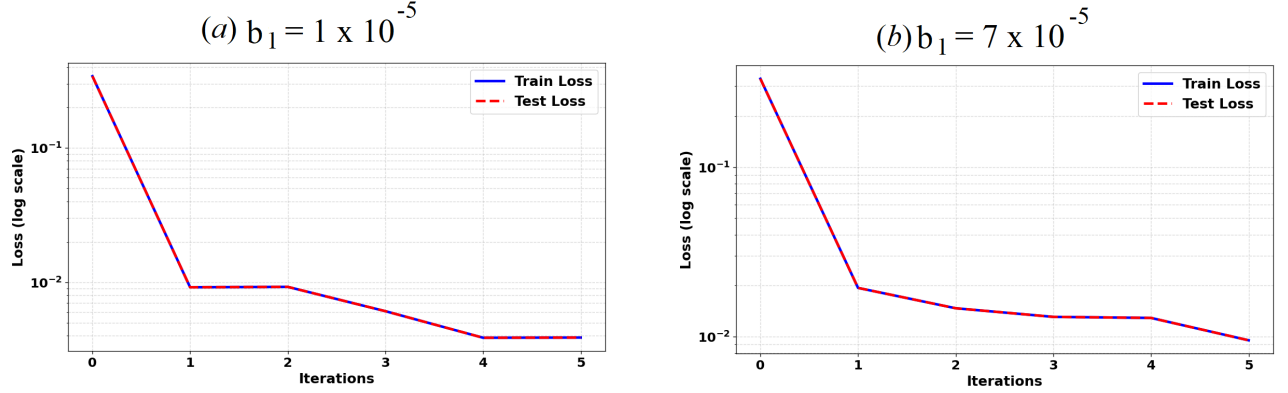


Figure 11: gaussian

4.10.1 Spatiotemporal Evolution of the Wave Profile

The spatiotemporal evolution of the wave amplitude $|\phi(x, t)|^2$ under the Gaussian initial condition is illustrated through three-dimensional surface plots (Figures 10(a)–10(c)). The results show the formation of a highly localized wave structure centered at $x = 0$, resembling a bright soliton.

For all values of the inverse nonlinear interaction parameter b_1 , the solution maintains a single dominant peak with rapid decay away from the center. The amplitude of the wave is highest at the center and decreases symmetrically along the spatial domain.

For $b_1 = 1 \times 10^{-5}$, the wave exhibits a sharp and highly localized peak with maximum amplitude. As b_1 increases to 4×10^{-5} and 7×10^{-5} , the peak amplitude decreases while the wave profile becomes broader. This indicates a reduction in localization as the inverse nonlinear interaction strengthens.

4.10.2 Contour Representation of Wave Dynamics

The contour plots (Figures 10(d)–10(f)) provide a two-dimensional visualization of the wave evolution in the (x, t) domain. A distinct vertical bright band is observed at $x \approx 0$, representing the localized high-amplitude region.

The vertical alignment of the contour indicates that the wave remains stationary over time, with no spatial propagation. The width of the bright band increases with larger values of b_1 , indicating a broader distribution of wave energy.

For smaller values of b_1 , the contour band is narrower and more intense, confirming stronger localization and higher peak amplitude.

4.10.3 Cross-Sectional Profile Analysis

The one-dimensional cross-sectional profiles (Figure 10(g)) of $|\phi(x, t)|^2$ further illustrate the effect of the parameter b_1 . All profiles show a single symmetric peak centered at $x = 0$.

For $b_1 = 1 \times 10^{-5}$, the peak is narrow and reaches the highest amplitude, indicating strong localization. As b_1 increases, the peak becomes wider and its amplitude decreases. Additionally, slight side lobes are observed for smaller values of b_1 , indicating stronger nonlinear effects.

This behavior demonstrates that increasing b_1 reduces the confinement of the wave and spreads the energy over a wider spatial region.

4.10.4 Effect of Parameter b_1

The inverse nonlinear interaction parameter b_1 significantly influences the amplitude and localization of the wave. The results indicate that:

- Smaller values of b_1 enhance nonlinear focusing, producing sharp and highly localized peaks.
- Larger values of b_1 reduce localization, resulting in broader wave profiles with lower peak amplitude.

4.10.5 Stability of the Solution

The numerical simulations confirm that the wave structure remains stable for all considered values of b_1 . No numerical instability or wave distortion is observed over the temporal domain. The consistent agreement between surface plots, contour plots, and cross-sectional profiles demonstrates the robustness of the PINN framework.

4.11 Training Performance Analysis for Gaussian Initial Condition

4.11.1 Convergence Behavior

Figure 11(a)–11(b) illustrates the convergence behavior of the Physics-Informed Neural Network (PINN) model for the Gaussian initial condition under different values of the inverse nonlinear interaction parameter b_1 .

For both $b_1 = 1 \times 10^{-5}$ and $b_1 = 7 \times 10^{-5}$, the training and test losses decrease rapidly during the initial iterations, indicating that the model effectively captures the localized wave structure. The loss drops sharply after the first iteration and continues to decrease gradually, demonstrating stable convergence.

4.11.2 Training and Test Loss Comparison

The training and test loss curves shown in Figure 11(a)–11(b) almost completely overlap for both parameter values, indicating strong generalization capability and the absence of overfitting. This behavior confirms that the PINN framework successfully enforces the governing physical constraints.

For $b_1 = 1 \times 10^{-5}$, the loss decreases quickly and stabilizes at a low value with minimal fluctuations. Similarly, for $b_1 = 7 \times 10^{-5}$, the loss follows a smooth decreasing trend, with slightly higher values compared to the smaller b_1 case.

4.11.3 Effect of Parameter b_1 on Convergence

The parameter b_1 influences the convergence characteristics of the training process. For the smaller value $b_1 = 1 \times 10^{-5}$, the convergence is slightly faster and reaches a lower loss value, indicating easier learning of strongly localized wave structures.

For the larger value $b_1 = 7 \times 10^{-5}$, the convergence remains stable but occurs at a slightly higher loss level, reflecting increased difficulty in approximating broader wave profiles.

4.11.4 Stability and Model Robustness

In both cases, the loss values decrease smoothly without significant oscillations or divergence, demonstrating the numerical stability of the training process. The rapid convergence within a few iterations highlights the efficiency of the PINN approach.

The close agreement between training and test losses further validates the robustness of the model in accurately approximating nonlinear wave dynamics under the Gaussian initial condition.

4.11.5 Conclusion

The results presented in Figures 10 and 11 demonstrate that the Gaussian initial condition produces stable and localized wave structures similar to bright solitons. The parameter b_1 plays a crucial role in controlling the localization and amplitude of the wave. Lower values of b_1 result in sharper and higher peaks, while larger values lead to broader and smoother profiles. These results confirm the effectiveness of the PINN approach in accurately capturing localized nonlinear wave dynamics.

4.12 Cosine initial condition

Using the PINN framework in DeepXDE, a numerical solution was found for the nonlinear Schrödinger-type system. It was determined that the time dimension was $t \in [0, 2]$, whereas the space dimension was $x \in [-10, 10]$. In the beginning of this experiment, the cosine function existed as the initial state.

$$u(x, 0) = \cos(x) \tag{15}$$

The value of the imaginary component was first set to zero ($v(x, 0) = 0$), which was the starting value. $a = 1.0$, $b_2 = -1.0$, and $b_3 = 0.0$ were the values that were used to establish the parameters. The inverse nonlinear interaction parameter b_1 was evaluated at three different values: 0.0001 , 1×10^{-5} , and 0.001 . This was done in order to determine the effect that it has on the evolution of periodic waves.

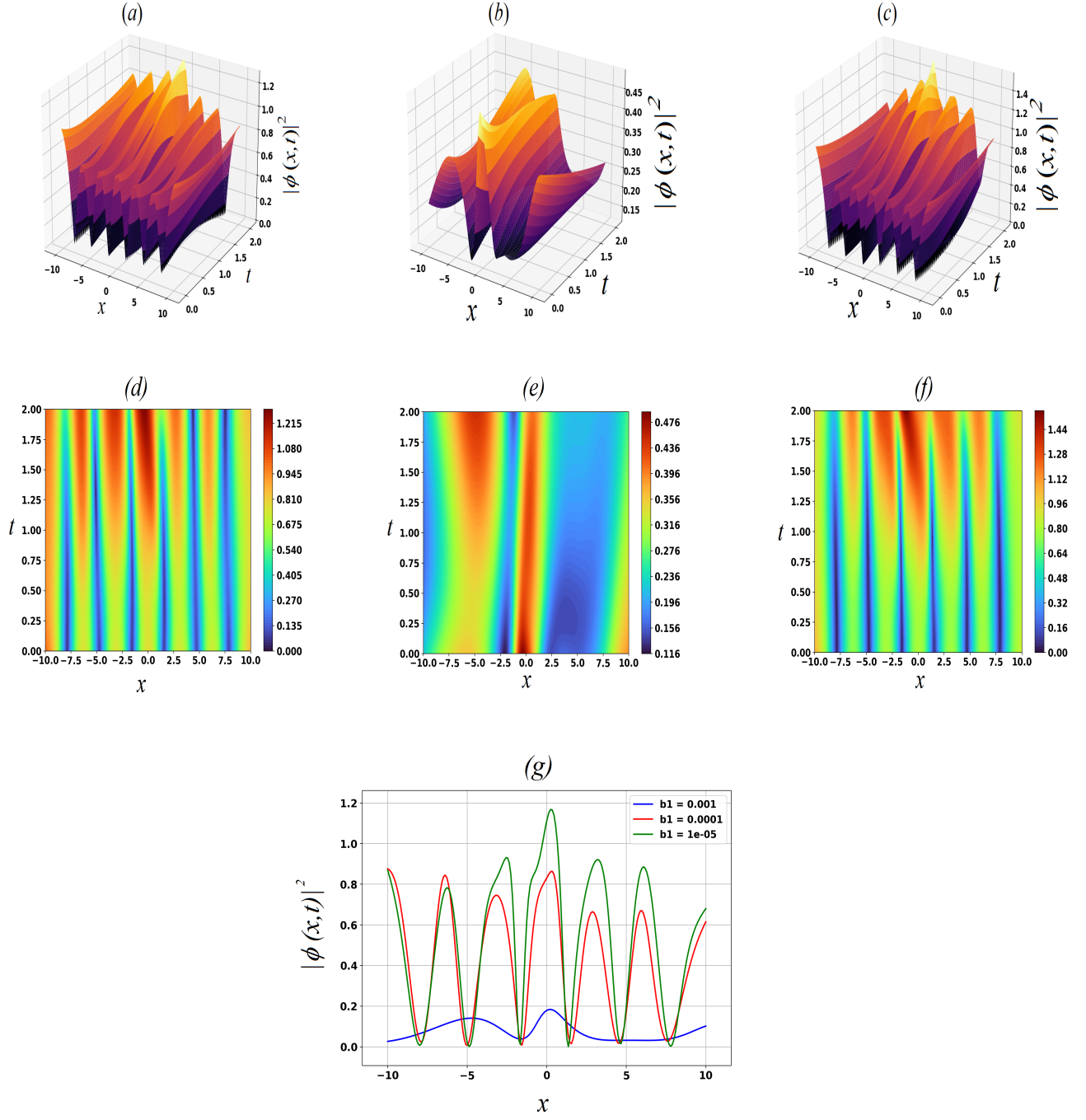


Figure 12:

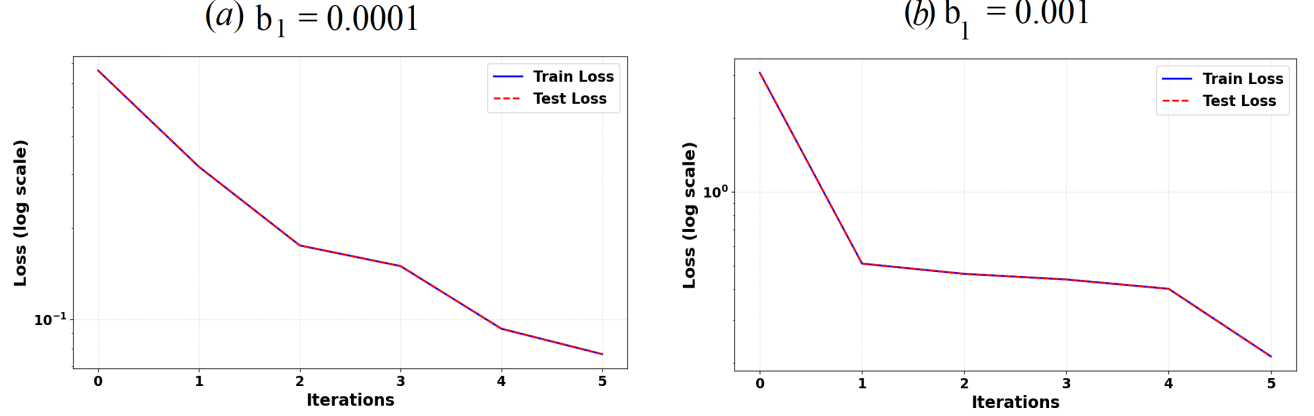


Figure 13: cos

4.12.1 Spatiotemporal Evolution of Periodic Waves

The spatiotemporal evolution of the wave amplitude $|\phi(x, t)|^2$ under the cosine initial condition is illustrated through three-dimensional surface plots (Figures 12(a)–12(c)). The results exhibit periodic wave structures distributed across the spatial domain, consistent with the nature of the cosine function.

For all values of the inverse nonlinear interaction parameter b_1 , the wave maintains a periodic structure with alternating peaks and troughs. The oscillatory pattern persists over time, indicating stable nonlinear wave behavior.

For $b_1 = 0.001$, the wave exhibits lower amplitude oscillations with smoother peaks. As b_1 decreases to 0.0001 and 10^{-5} , the amplitude of oscillations increases, and the peaks become sharper. This demonstrates the enhancement of nonlinear effects for smaller values of b_1 .

4.12.2 Contour Representation of Wave Dynamics

The contour plots (Figures 12(d)–12(f)) provide a two-dimensional visualization of the wave evolution in the (x, t) domain. Alternating vertical bands of high and low amplitude are observed, corresponding to the periodic peaks and troughs of the cosine wave.

The vertical alignment of the contour bands indicates that the wave remains stationary in space and does not propagate over time. The intensity of the bands increases as b_1 decreases, indicating stronger nonlinear amplification.

For larger values of b_1 , the contour bands are smoother and less pronounced, while smaller values result in sharper and more distinct periodic patterns.

4.12.3 Cross-Sectional Profile Analysis

The one-dimensional cross-sectional profiles (Figure 12(g)) of $|\phi(x, t)|^2$ confirm the periodic nature of the solution. The profiles exhibit regular oscillations across the spatial domain.

For $b_1 = 0.001$, the oscillations have lower amplitude and appear smoother. As b_1 decreases, the amplitude increases significantly, and the peaks become sharper with deeper

troughs. The smallest value, $b_1 = 10^{-5}$, produces the most pronounced oscillatory behavior.

This indicates that decreasing b_1 enhances nonlinear amplification and strengthens the periodic wave structure.

4.12.4 Effect of Parameter b_1

The inverse nonlinear interaction parameter b_1 plays a crucial role in controlling the amplitude and sharpness of the periodic waves. The results indicate that:

- Larger values of b_1 suppress oscillations, resulting in smoother wave profiles.
- Smaller values of b_1 enhance nonlinear effects, leading to higher amplitude and sharper periodic structures.

4.12.5 Stability of the Solution

The numerical simulations show that the periodic wave structures remain stable for all considered values of b_1 . No numerical instability or wave distortion is observed across the temporal domain. The agreement between surface plots, contour plots, and cross-sectional profiles confirms the robustness of the PINN framework.

4.13 Training Performance Analysis for Cosine Initial Condition

4.13.1 Convergence Behavior

Figure 13(a)–13(b) presents the convergence behavior of the Physics-Informed Neural Network (PINN) model for the cosine initial condition under different values of the inverse nonlinear interaction parameter b_1 .

For both $b_1 = 0.0001$ and $b_1 = 0.001$, the training and test losses decrease steadily over iterations, indicating that the model effectively learns the periodic wave structure. The loss shows a consistent downward trend without abrupt fluctuations, demonstrating stable convergence.

4.13.2 Training and Test Loss Comparison

The training and test loss curves shown in Figure 13(a)–13(b) closely overlap for both parameter values, indicating strong generalization and the absence of overfitting. This confirms that the PINN framework successfully incorporates the governing physical constraints of the nonlinear system.

For $b_1 = 0.0001$, the loss decreases smoothly with a gradual reduction across iterations, reaching a relatively low value. Similarly, for $b_1 = 0.001$, the loss follows a stable decreasing trend, although the initial loss values are slightly higher due to stronger nonlinear interactions.

4.13.3 Effect of Parameter b_1 on Convergence

The parameter b_1 influences the convergence characteristics of the training process. For $b_1 = 0.0001$, the convergence is smooth and monotonic, indicating efficient learning of the periodic wave dynamics.

For $b_1 = 0.001$, the convergence remains stable but occurs with slightly higher loss values in the early iterations, reflecting the increased complexity in capturing stronger nonlinear effects.

4.13.4 Stability and Model Robustness

In both cases, the loss decreases consistently without significant oscillations or divergence, demonstrating the numerical stability of the training process. The rapid reduction in loss within a few iterations highlights the efficiency of the PINN approach.

The close agreement between training and test losses further validates the robustness of the model in accurately approximating nonlinear wave behavior under the cosine initial condition.

5 Conclusion

In this study, a Physics-Informed Neural Network (PINN) framework has been successfully applied to solve a nonlinear Schrödinger-type equation incorporating anticubic nonlinearity. The model effectively captures the complex spatiotemporal dynamics of nonlinear wave propagation in optical fiber systems.

The numerical results obtained for various initial conditions, including hyperbolic tangent, sine, cosine, Gaussian, super-Gaussian, and hyperbolic secant functions, demonstrate the versatility of the PINN approach in modeling different wave structures such as kink waves, periodic waves, and localized soliton-like pulses. The model accurately reproduces both localized and oscillatory behaviors while maintaining stability across the computational domain.

A key finding of this study is the significant influence of the inverse nonlinear interaction parameter b_1 on wave dynamics. For localized initial conditions (Gaussian, super-Gaussian, and sech), smaller values of b_1 enhance nonlinear focusing, resulting in sharper and highly localized soliton structures. Conversely, larger values of b_1 produce broader wave profiles with reduced peak amplitude. For periodic initial conditions (sine and cosine), decreasing b_1 leads to stronger nonlinear amplification, increasing the amplitude and sharpness of oscillations.

The training performance analysis further confirms the robustness of the proposed PINN model. The rapid convergence of both training and test losses, along with their close agreement, indicates strong generalization capability and absence of overfitting. The model demonstrates stable and efficient learning behavior across all initial conditions and parameter settings.

Overall, the results validate that the PINN framework provides an accurate, stable, and computationally efficient approach for solving nonlinear partial differential equations with higher-order nonlinearities. The ability of PINNs to handle both localized and periodic wave structures highlights their potential for studying complex nonlinear optical systems.

Future work may focus on extending the model to higher-dimensional systems, incorporating additional physical effects such as higher-order dispersion and external perturbations, and exploring real-world optical communication applications.

Declarations

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Author's contributions

The authors declare that the study was realized in collaboration with the same responsibility. All authors read and approved the manuscript.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have neither financial nor conflict interest.

Ethical approval

Not applicable

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